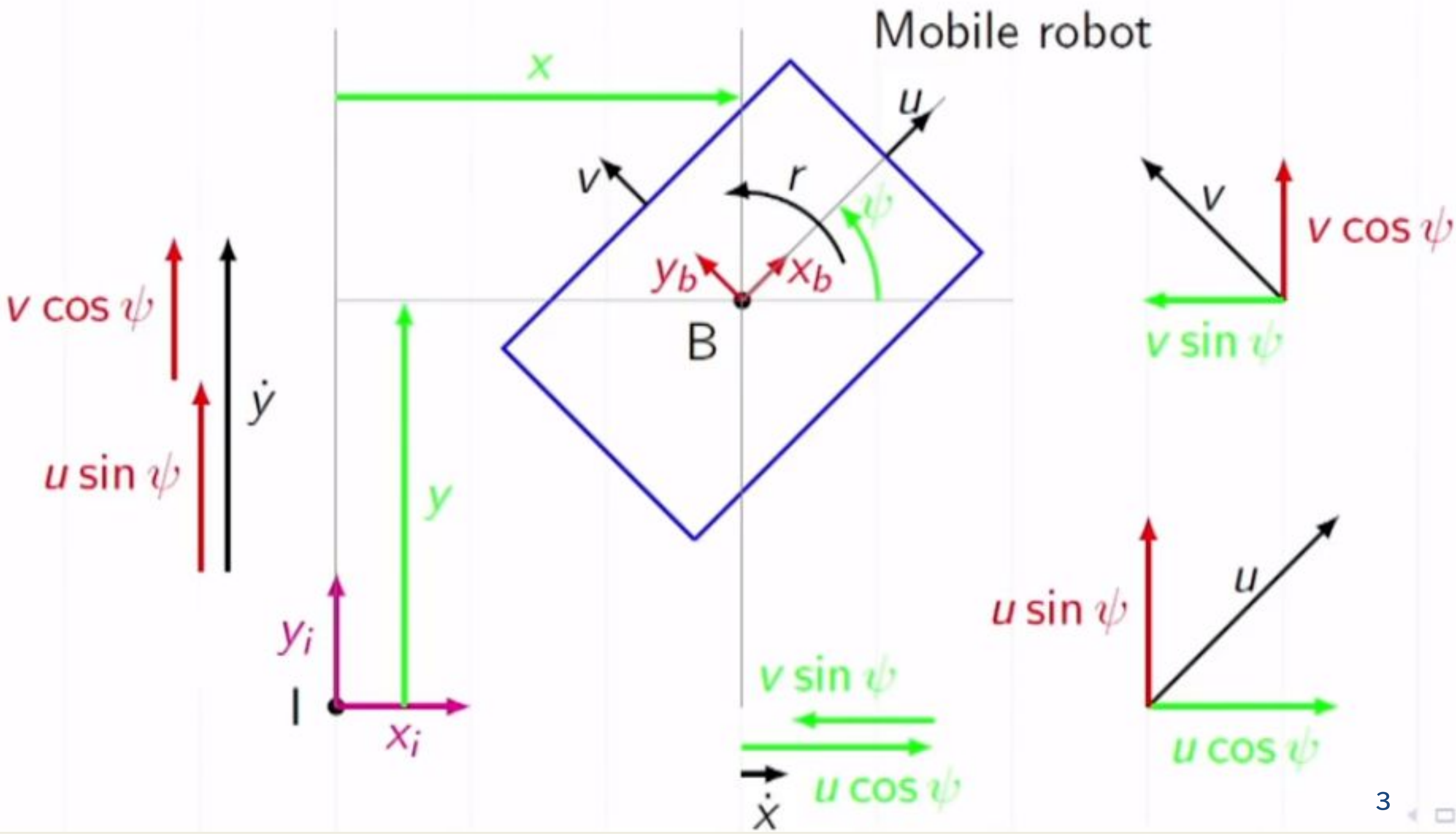
DEGREE OF MANEUVERABILITY AND TYPES OF WHEELS

WHAT ARE WE GOING TO LEARN?

- Recap: Differential Kinematics
  - Forward differential kinematics
  - Inverse differential kinematics
- Degree of Maneuverability
- Types of Wheels


Mobile robot
x
u
v
r
ψ
y_b
x_b
B
v
v cos ψ
v sin ψ
v cos ψ
ẏ
u sin ψ
y
y_i
I
x_i
u
u sin ψ
u cos ψ
v sin ψ
u cos ψ
ẋ

x: Forward displacement of the mobile robot w.r.t. I

y: Lateral displacement of the mobile robot w.r.t. I

ψ: Angular displacement of the mobile robot w.r.t. I

u: Forward velocity of the mobile robot w.r.t. B

v: Lateral velocity of the mobile robot w.r.t. B

r: Angular velocity of the mobile robot w.r.t. B

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} u\cos\psi - v\sin\psi \\ u\sin\psi + v\cos\psi \\ r \end{bmatrix}$$

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} u\cos\psi - v\sin\psi \\ u\sin\psi + v\cos\psi \\ r \end{bmatrix} = \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix}$$

$$\dot{\boldsymbol{\eta}} = \mathbf{J}(\psi)\,\boldsymbol{\zeta}$$

It describes the relation between the velocity input commands (ζ) and the derivatives of generalized coordinates ($\dot{\eta}$).

- J is the jacobian matrix or velocity transformation matrix

FORWARD DIFFERENTIAL KINEMATICS

- For given input velocity commands, finding the derivatives of generalized coordinates

$$\dot{\eta} = \mathbf{J}(\psi)\,\zeta$$

- Simulating or analysing the system in velocity level

INVERSE DIFFERENTIAL KINEMATICS

- For the desired derivatives of generalized coordinates (or given position trajectory), finding the corresponding velocity input commands.

$$\zeta = \mathbf{J}^{-1}(\psi)\,\dot{\boldsymbol{\eta}}$$

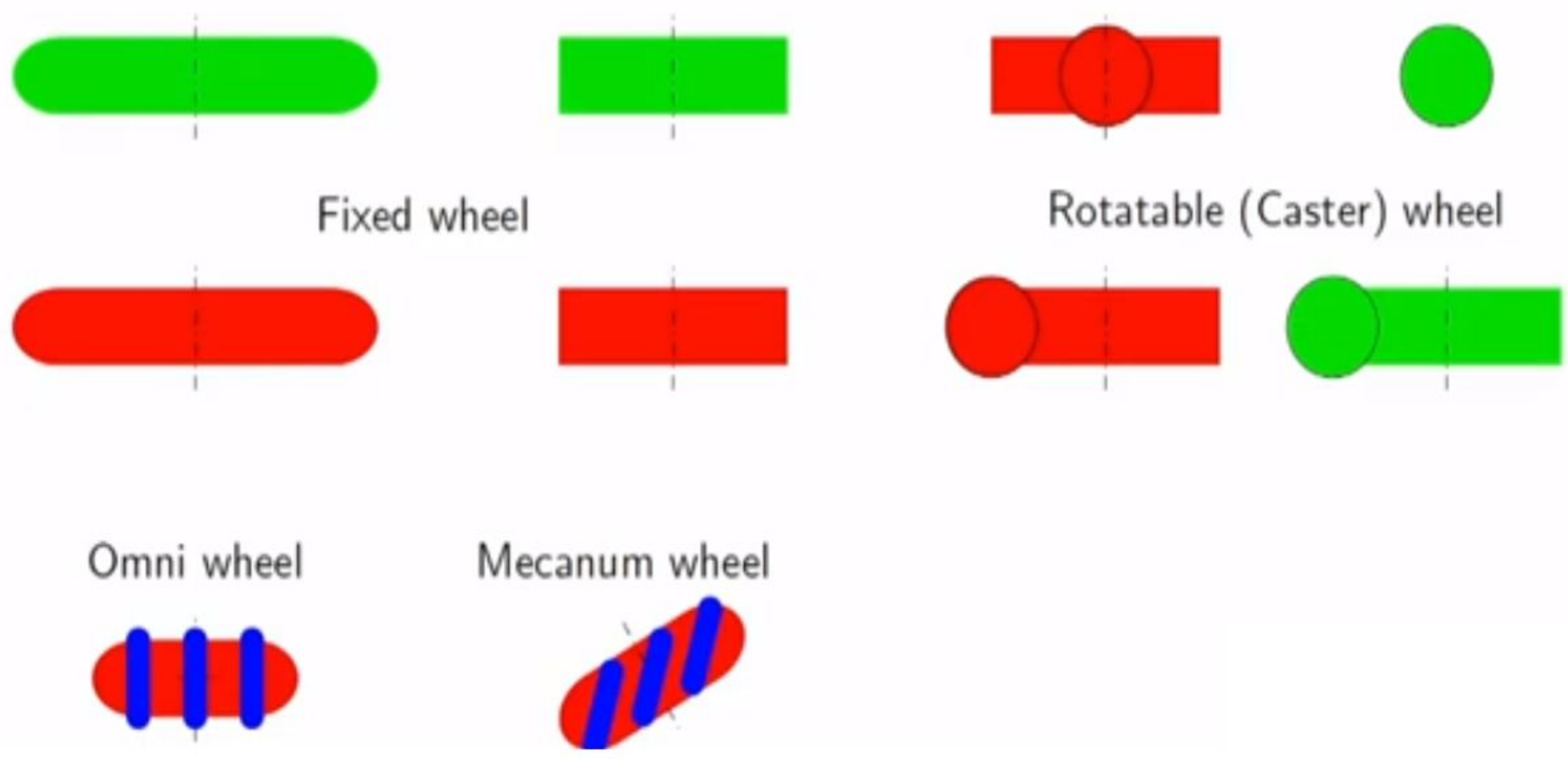
- Controlling the system in velocity level

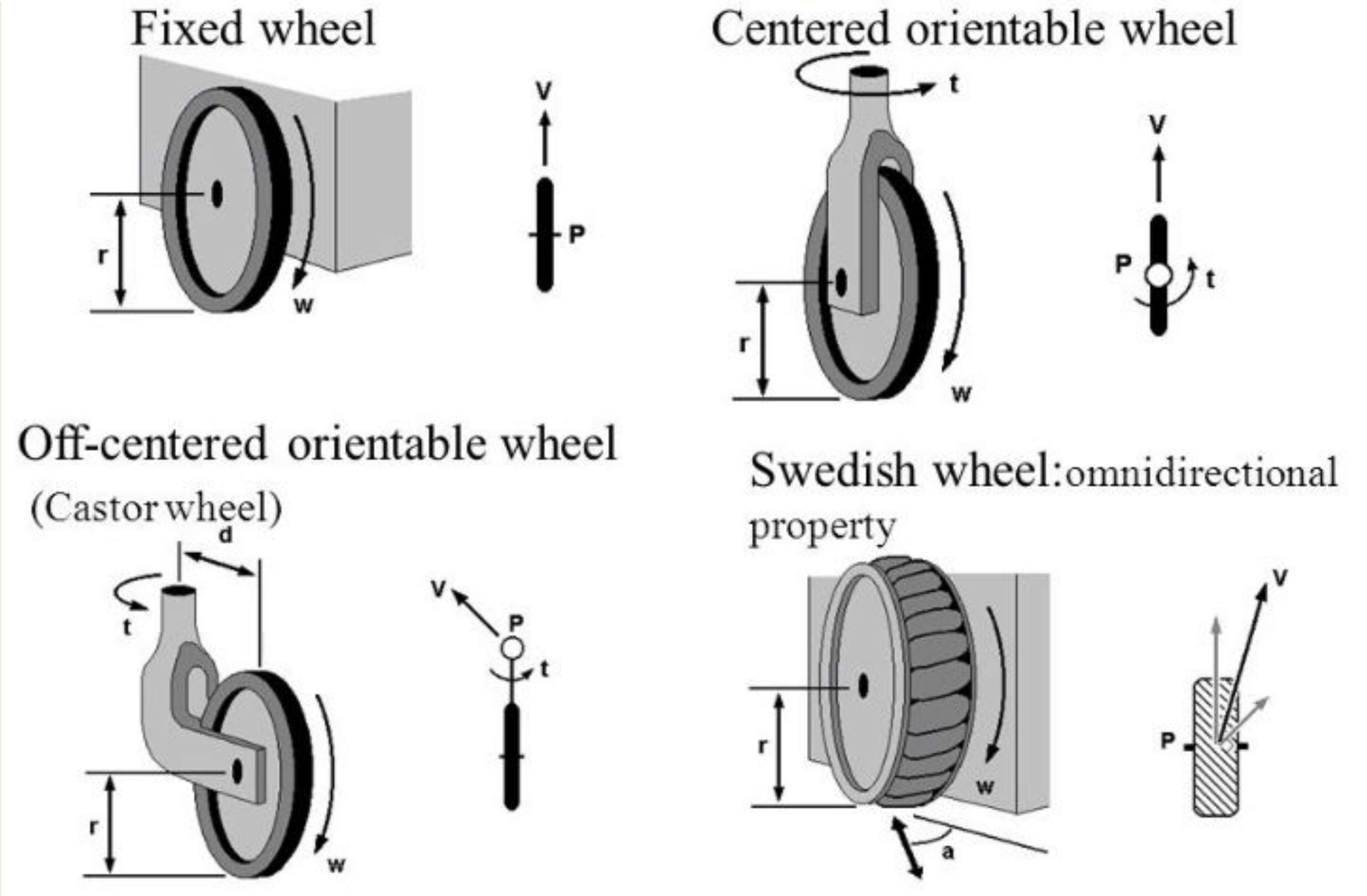
DEGREE OF MANEUVERABILITY

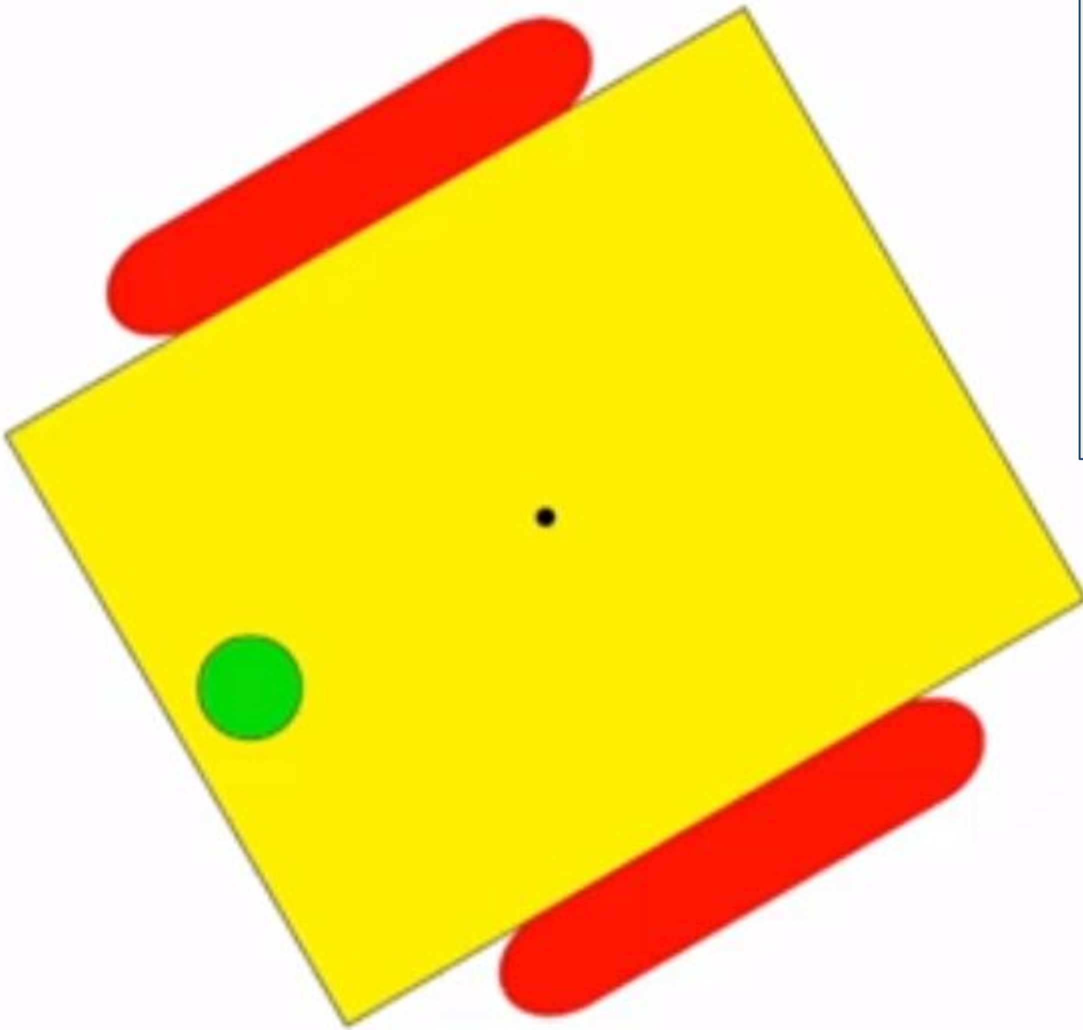
- Sum of degree of mobility and steerability
- It includes both the degree of freedom that the robot manipulates directly through wheel velocity and the degree of freedom that it directly manipulates by changing the steering configuration and moving.
- In other words, it is the total number of controllable inputs.
- Degree of maneuverability depends on the kinematic configuration and actuator arrangement of the mobile robot.

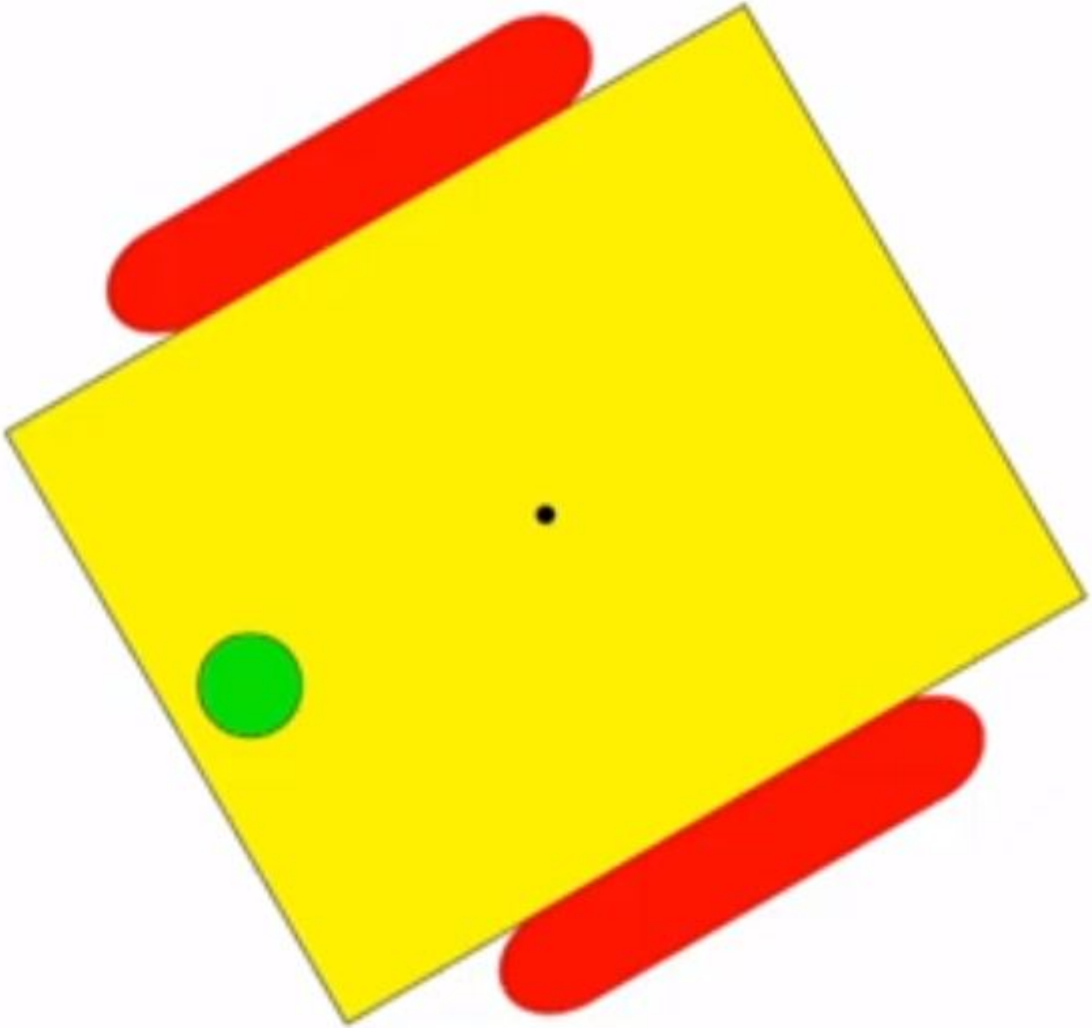
- Degree of maneuverability = Degree of mobility + Degree of steerability
-

$$\delta_M = \delta_m + \delta_s$$


Fixed wheel
Rotatable (Caster) wheel
Omni wheel
Mecanum wheel


Fixed wheel
V
P
r
w
Centered orientable wheel
t
V
P
t
r
w
Off-centered orientable wheel
(Castor wheel)
d
t
V
P
t
r
w
Swedish wheel: omnidirectional
property
V
P
r
w
a

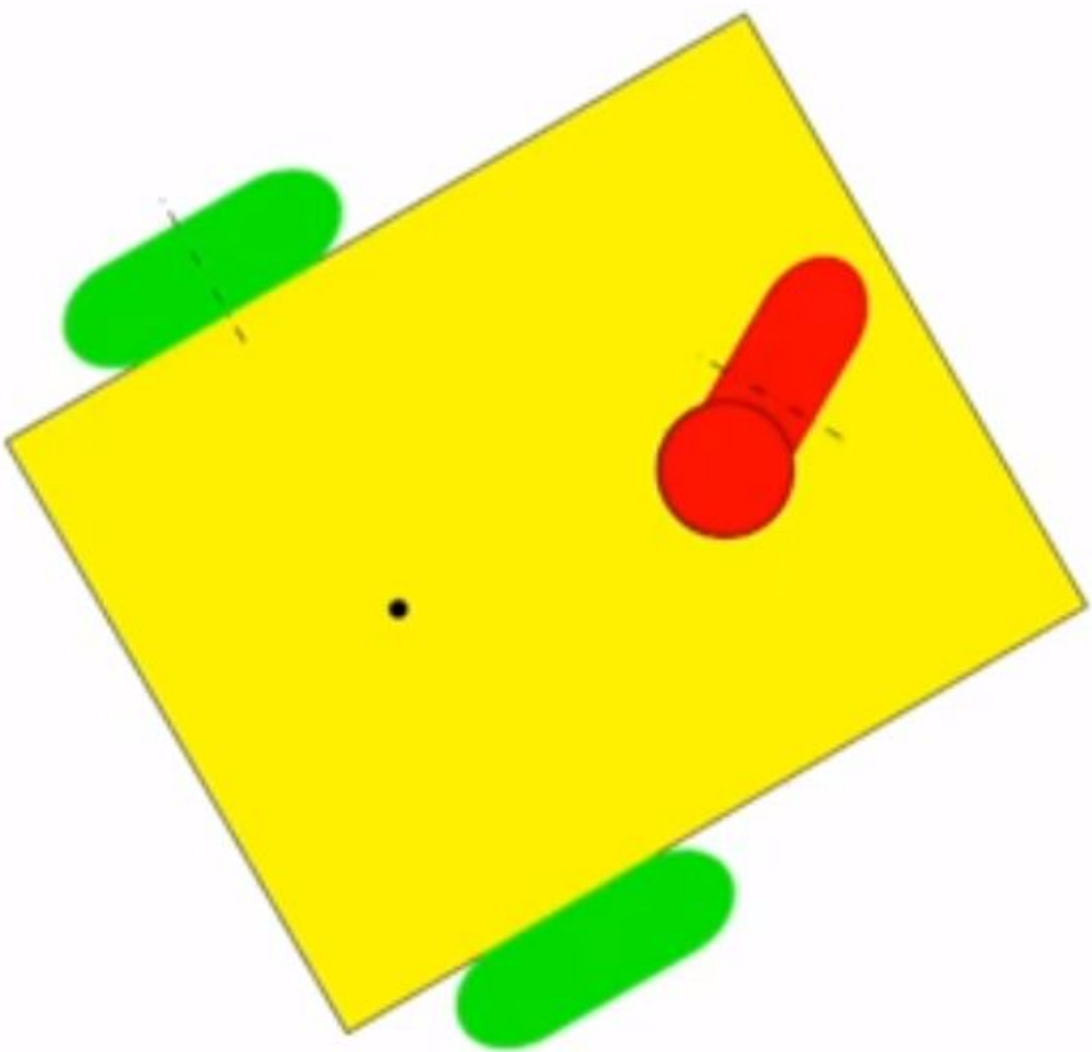


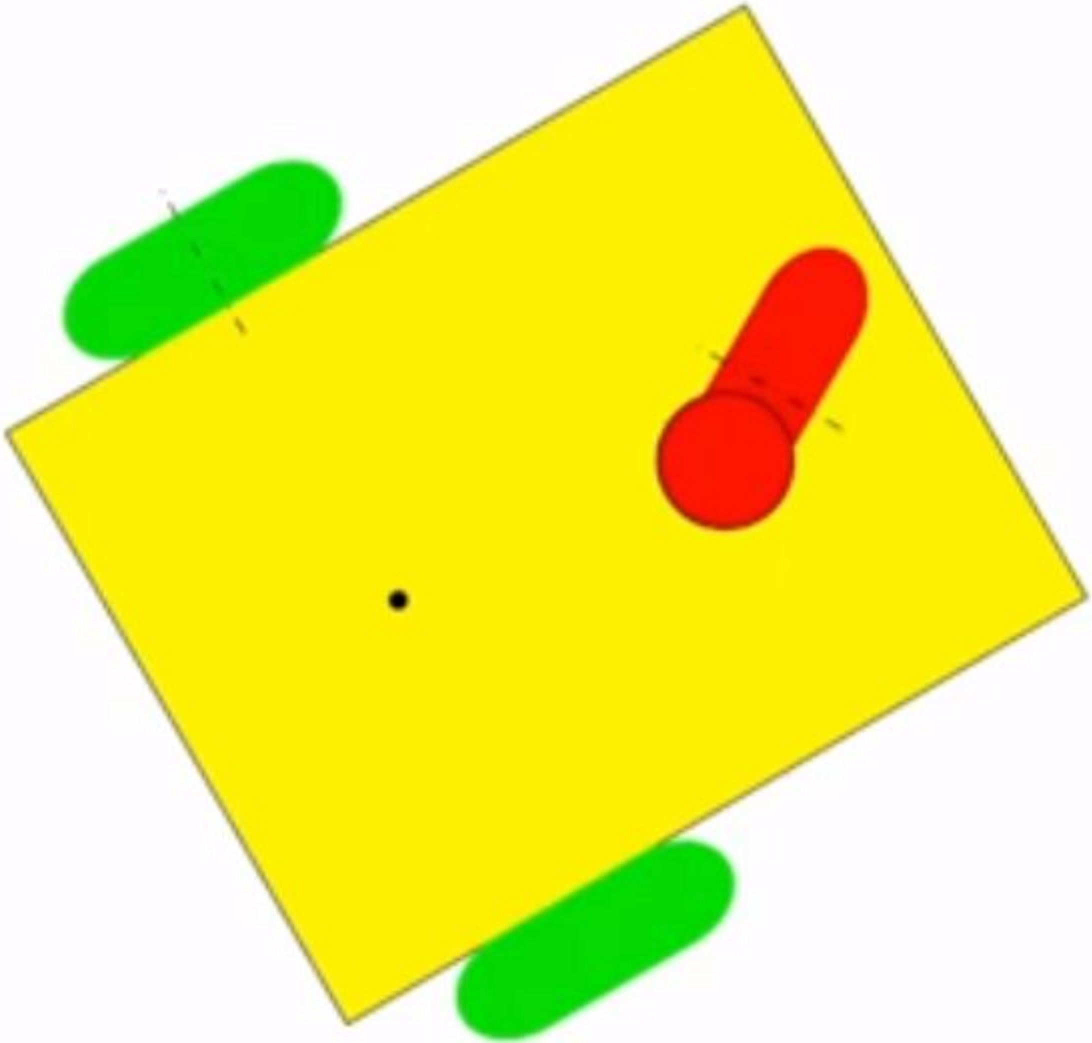


$$\delta_m = 2$$

$$\delta_s = 0$$

$$\delta_M = \delta_m + \delta_s = 2$$

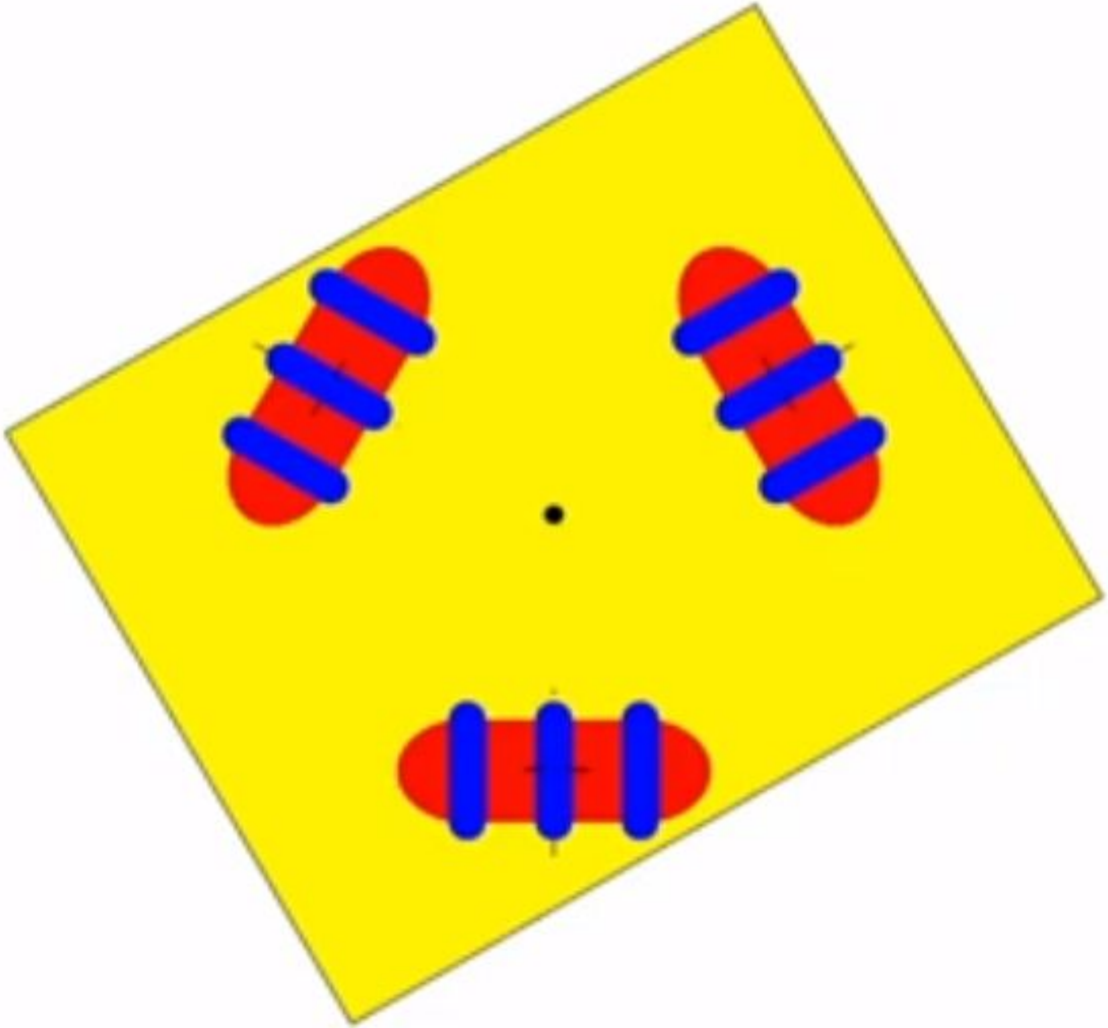


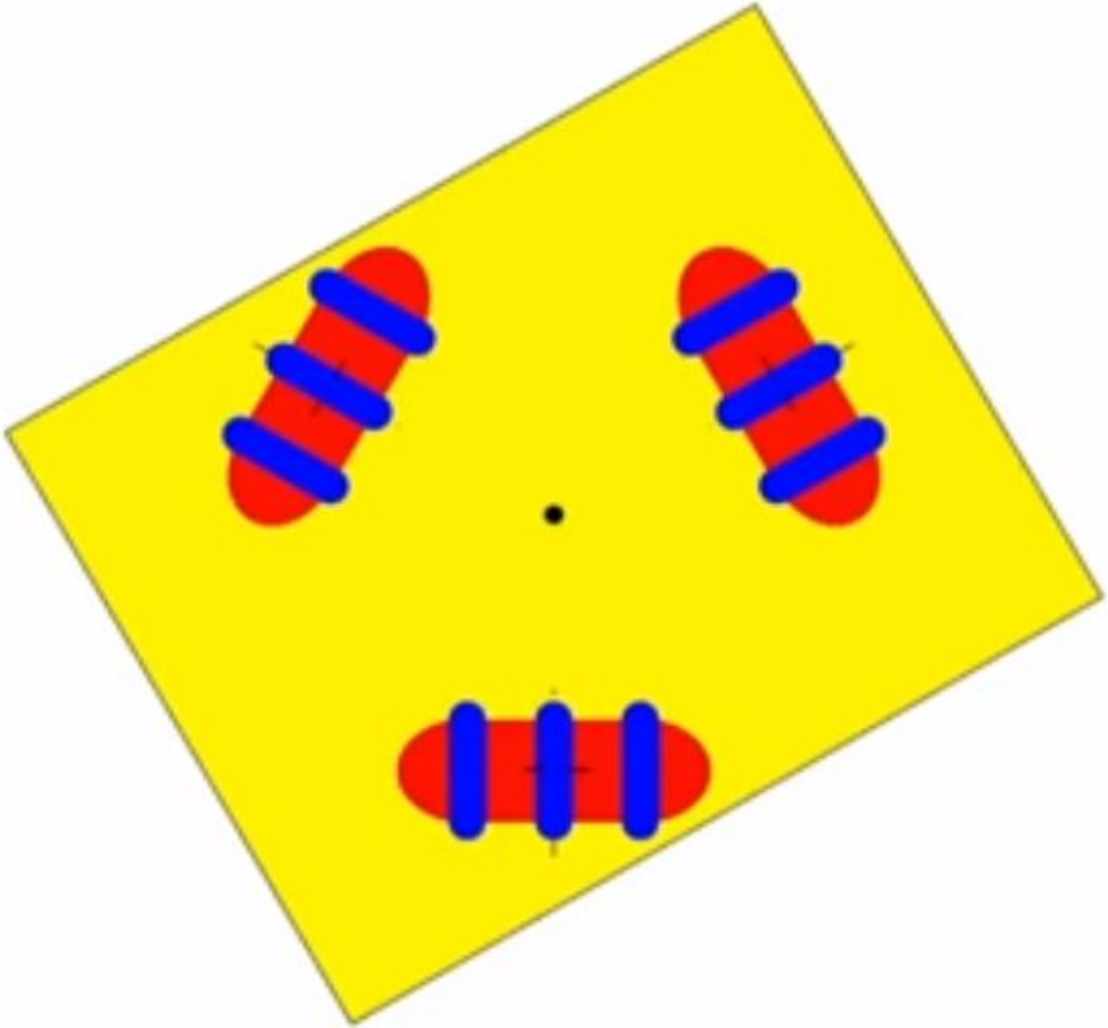


$$\delta_m = 1$$

$$\delta_s = 1$$

$$\delta_M = \delta_m + \delta_s = 2$$

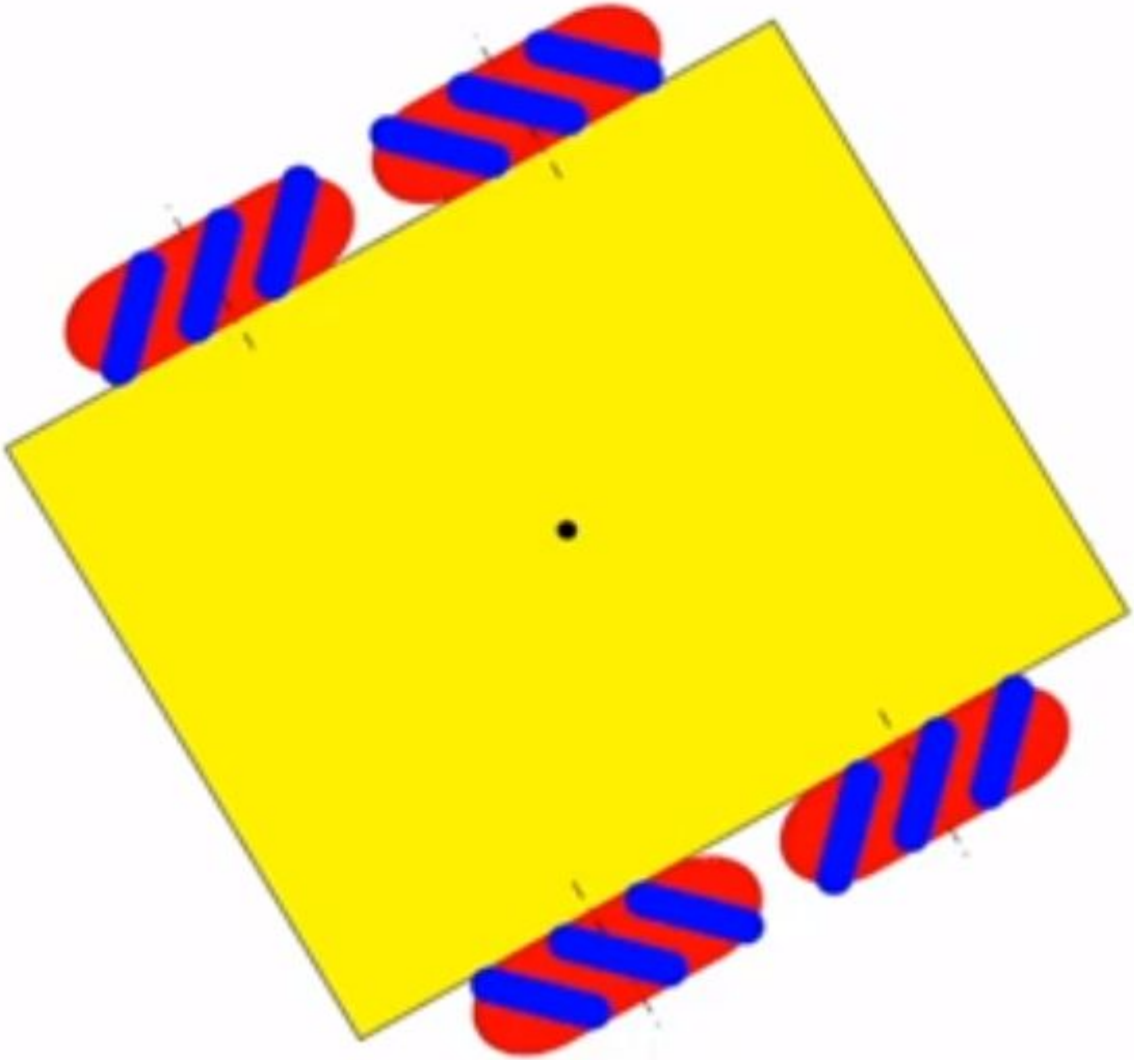


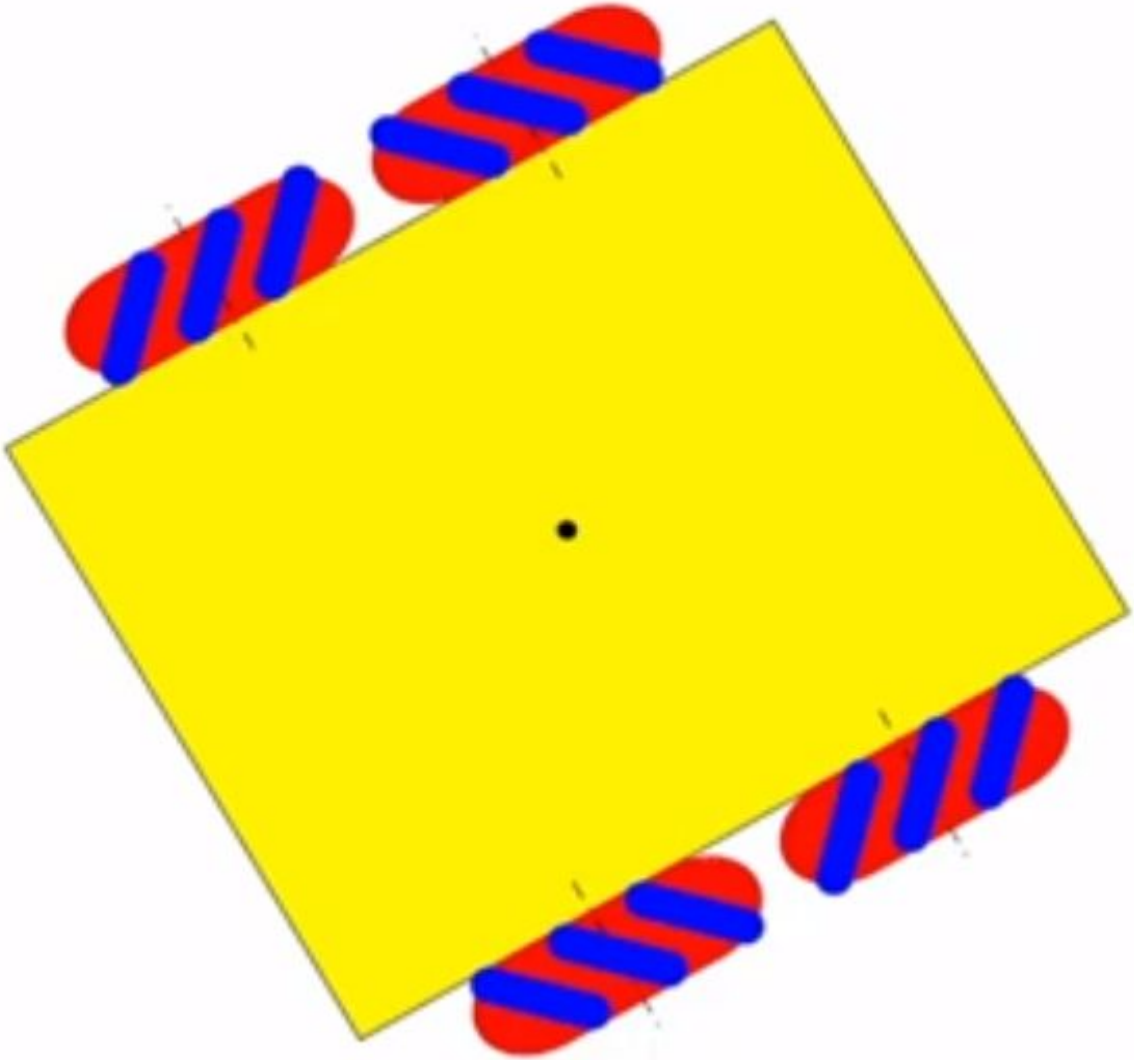


$$\delta_m = 3$$

$$\delta_s = 0$$

$$\delta_M = \delta_m + \delta_s = 3$$

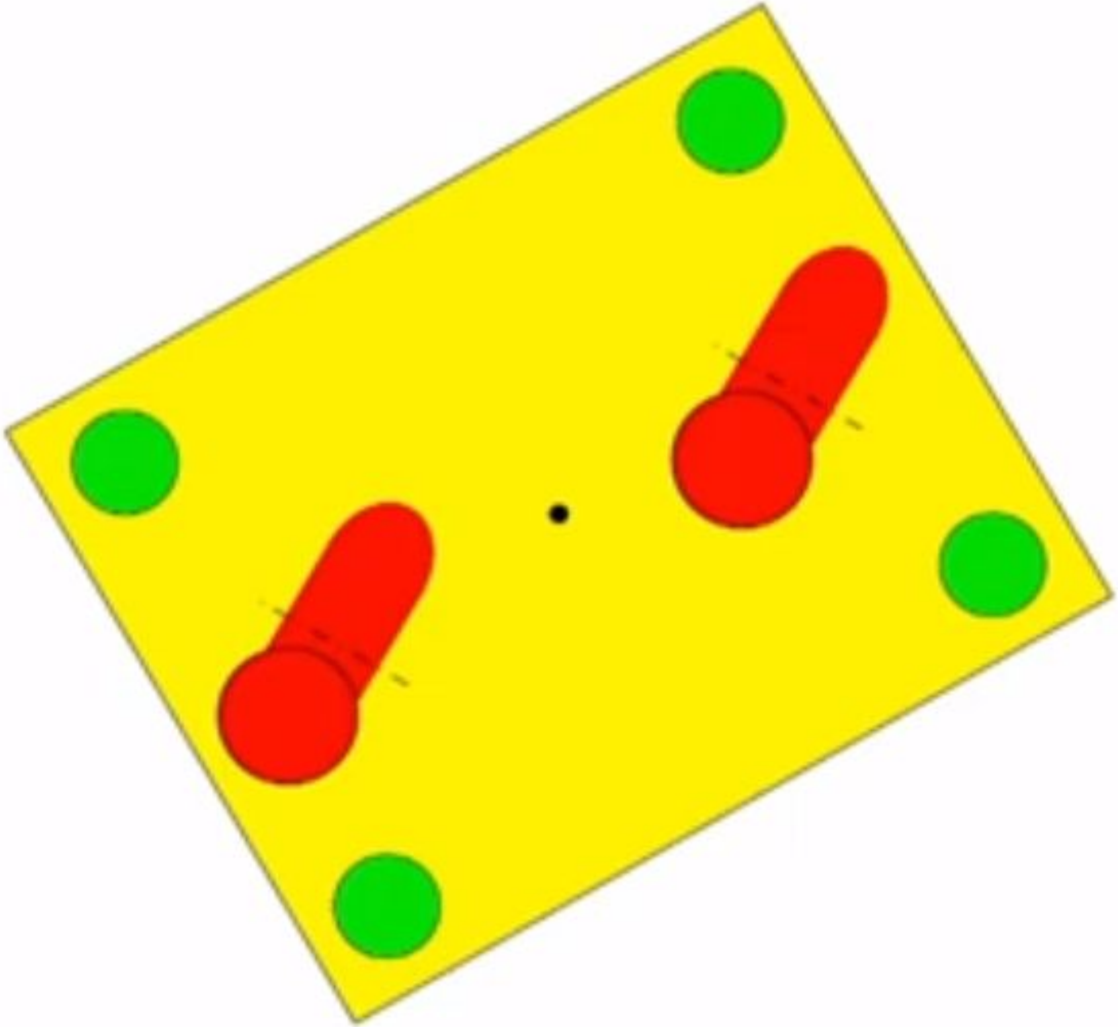


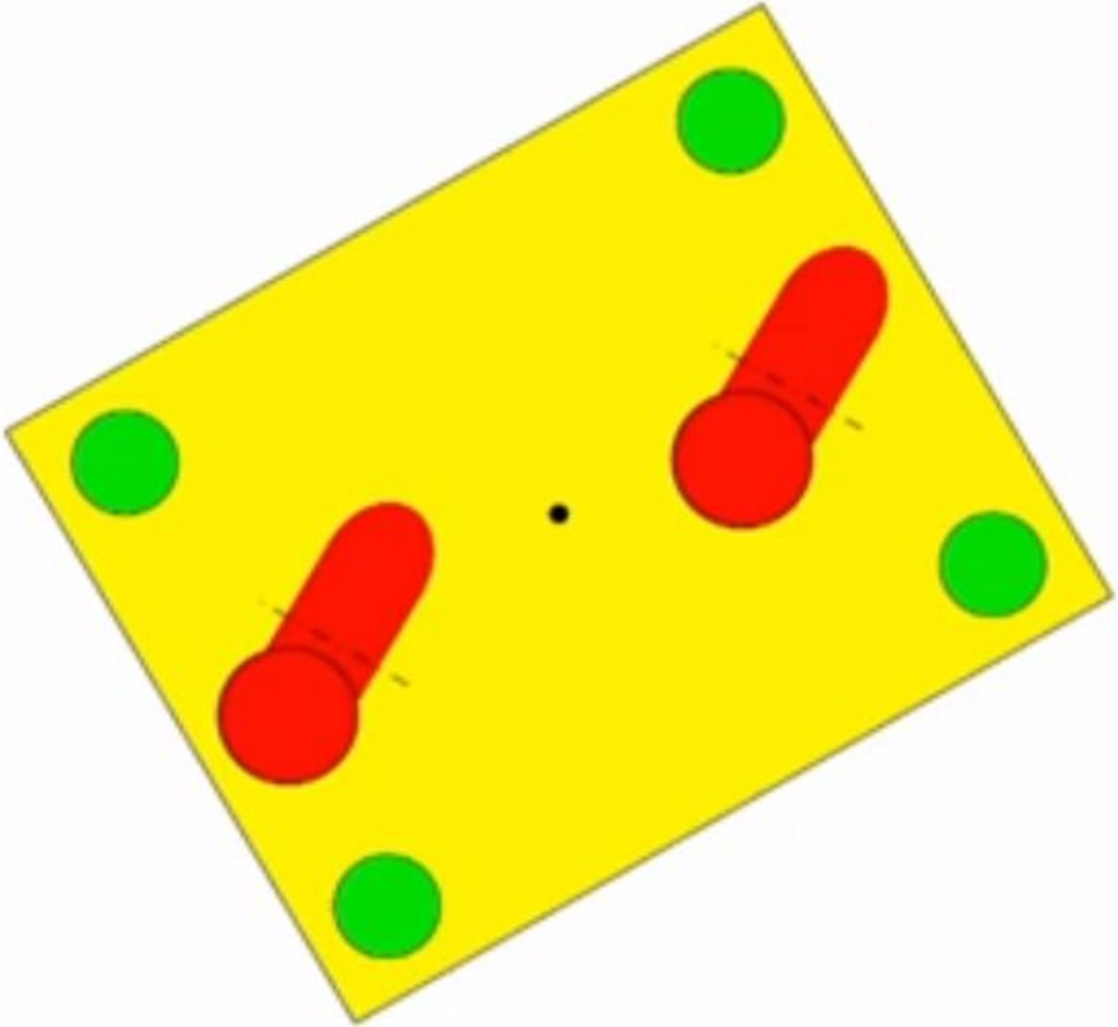


$$\delta_m = 4$$

$$\delta_s = 0$$

$$\delta_M = \delta_m + \delta_s = 4$$

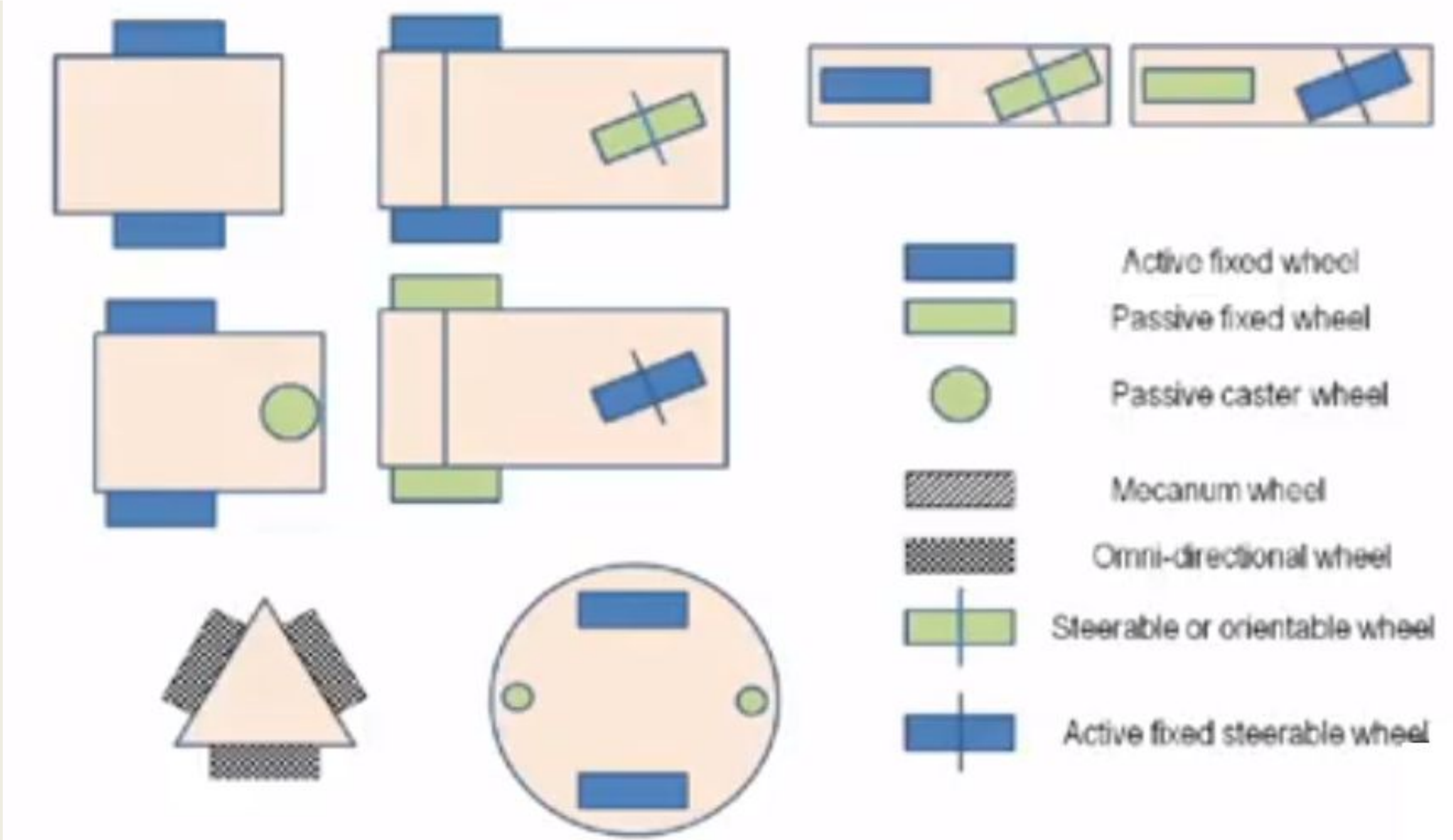




$$\delta_m = 2$$

$$\delta_s = 2$$

$$\delta_M = \delta_m + \delta_s = 4$$



Active fixed wheel
Passive fixed wheel
Passive caster wheel
Mecanum wheel
Omni-directional wheel
Steerable or orientable wheel
Active fixed steerable wheel